Team Greenlight

DSN x BCT LLM Agent Hackathon 3.0

Project Overview — Plain English Edition

What We Built | How It Works | Why It Matters

1. What Is This Project About?

We built an AI system that reads thousands of product and restaurant reviews, learns each person's taste, and then does two powerful things: (1) writes a new review the way that specific person would write it, and (2) recommends products that person would actually enjoy — all in Nigerian English.

Think of it like this: if you gave our AI your name, it would study all your past reviews, figure out whether you love spicy food or prefer mild, whether you write long detailed opinions or short punchy ones, and whether you tend to rate things 4 stars or 2 stars. Then it could write a review *as you* for any new item, or suggest products you'd probably love.

This is called **User Modelling and Personalised Recommendation** — two of the most in-demand skills in AI right now, used by Netflix, Spotify, and Amazon every day.

2. The Two Things Our System Does

Task A — Writing Reviews in Your Style

You give the system a user ID and a product name. The AI looks up that user's history, understands their personality (Do they love things? Are they critical? Do they write essays or one-liners?), and generates a completely new review that sounds exactly like them — in Nigerian English with natural Pidgin phrases like *"I no go lie," "e dey sweet,"* and *"abeg."*

Task B — Recommending Things You'd Love

You give the system a user ID and it searches through nearly 10,000 real reviews to find items that match that person's taste. It doesn't just match keywords — it understands meaning. If you love 'bold flavours' it finds items described as 'intense' or 'rich' too. Then an AI reranks the results and explains each recommendation in Pidgin English.

Cold-Start (New Users)

What if someone is brand new with no history? Our system handles that too. You just describe what you want in plain English — *"healthy low-sugar office snacks"* — and it finds matching items without needing any past reviews.

3. Where Did the Data Come From?

We used three large public datasets totalling nearly 10,000 real human reviews:

- **Yelp** — restaurant and business reviews (5,000 reviews)
- **Amazon Grocery** — food product reviews (3,000 reviews)
- **Amazon Books** — book reviews used in place of Goodreads, which is no longer publicly available (2,000 reviews)

From these we built profiles for 1,271 users — storing each person's average rating, writing style, typical review length, and most-used words. All of this is processed automatically by our data pipeline.

4. Why Nigerian English?

The hackathon specifically rewards cultural relevance for African markets. We localised the entire output layer to use natural Nigerian English and Pidgin expressions — not awkward translations, but the way Nigerians actually talk online and in reviews.

Phrases like *"I no go lie, this burger dey mad!"* or *"abeg, the service na wa o"* are baked into our AI prompts so every recommendation explanation and generated review feels local and authentic, not foreign.

5. How Well Does It Work?

Task A — Review Generation Results (16 users tested):

Metric	Score	What It Means
Text Similarity (ROUGE-L)	0.1357	Vocabulary overlap with real reviews
Rating Accuracy (RMSE)	0.87 stars	Average star prediction error
Style Match Rate	100%	User profile found every time
Avg Review Length	1,142 chars	Detailed, substantial reviews

Task B — Recommendation Results (20 users + 5 cold-start):

Metric	Score	What It Means
Hit Rate@5	0.6500 (65%)	Ground truth item in top 5 for 13 of 20 users
NDCG@5	0.2968	Relevant items appear near the top of rankings
Cold-start success	5/5 (100%)	All new users got relevant recommendations

Users evaluated	20/20	Full evaluation set completed

6. Technology (Simplified)

Here is what is powering everything under the hood, in plain terms:

- **Google Gemini 2.5 Flash** — the large language model that reads user profiles and writes reviews and recommendation explanations
- **FAISS** — a Facebook search engine that finds similar reviews at lightning speed (like Google but for meaning)
- **Sentence Transformers** — converts text into numbers so the search engine can compare meaning, not just words
- **FastAPI** — the web framework that serves everything as a professional API with two separate services
- **Docker** — packages everything into containers so the system runs identically anywhere

The entire system is containerised, documented, and ready to deploy. Both APIs have health checks, full error handling, and professional documentation built in.

7. Team Greenlight

We are competing in the DSN x BCT LLM Agent Hackathon 3.0 under the team name **Greenlight**. This document covers our submission for the User Modelling and Recommendation track. The full technical solution paper is available separately for judges and technical reviewers.

— Team Greenlight, May 2026